Research Analysis Report

Research Topic: Multi-Agent Systems

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'Multi-Agent Systems' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Consensus and Cooperation in Networked Multi-Agent Systems', 'year': 2007, 'summary': 'Abstract not available for analysis.'}

• {'title': 'Why Do Multi-Agent LLM Systems Fail?', 'year': 2025, 'summary': 'Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal. This gap highlights a critical need for a principled understanding of why MAS fail. Addressing this question requires systematic identification and analysis of failure patterns. We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks. MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems. To enable systematic classification of failures for MAST-Data, we build the first Multi-Agent System Failure Taxonomy (MAST). We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement (kappa = 0.88). This process identifies 14 unique modes, clustered into 3 categories: (i) system design issues, (ii) inter-agent misalignment, and (iii) task verification. To enable scalable annotation, we develop an LLM-as-a-Judge pipeline with high agreement with human annotations. We leverage MAST and MAST-Data to analyze failure patterns across models (GPT4, Claude 3, Qwen2.5, CodeLlama) and tasks (coding, math, general agent), demonstrating improvement headrooms from better MAS design. Our analysis provides insights revealing that identified failures require more sophisticated solutions, highlighting a clear'}

• {'title': 'Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems', 'year': 2025, 'summary': "Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive. In this paper, we propose and formulate a new research area: automated failure attribution for LLM multi-agent systems. To support this initiative, we introduce the Who&When; dataset, comprising extensive failure logs from 127 LLM multi-agent systems with fine-grained annotations linking failures to specific agents and decisive error steps. Using the Who&When;, we develop and evaluate three automated failure attribution methods, summarizing their corresponding pros and cons. The best method achieves

53.5% accuracy in identifying failure-responsible agents but only 14.2% in pinpointing failure steps, with some methods performing below random. Even SOTA reasoning models, such as OpenAI o1 and DeepSeek R1, fail to achieve practical usability. These results highlight the task's complexity and the need for further research in this area. Code and dataset are available at https://github.com/mingyin1/Agents_Failure_Attribution"}

• {'title': 'Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments', 'year': 2017, 'summary': 'We explore deep reinforcement learning methods for multi-agent domains. We begin by analyzing the difficulty of traditional algorithms in the multi-agent case: Q-learning is challenged by an inherent non-stationarity of the environment, while policy gradient suffers from a variance that increases as the number of agents grows. We then present an adaptation of actor-critic methods that considers action policies of other agents and is able to successfully learn policies that require complex multi-agent coordination. Additionally, we introduce a training regimen utilizing an ensemble of policies for each agent that leads to more robust multi-agent policies. We show the strength of our approach compared to existing methods in cooperative as well as competitive scenarios, where agent populations are able to discover various physical and informational coordination strategies.'}

• {'title': 'Red-Teaming LLM Multi-Agent Systems via Communication Attacks', 'year': 2025, 'summary': 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications. While the communication framework is crucial for agent coordination, it also introduces a critical yet unexplored security vulnerability. In this work, we introduce Agent-in-the-Middle (AiTM), a novel attack that exploits the fundamental communication mechanisms in LLM-MAS by intercepting and manipulating inter-agent messages. Unlike existing attacks that compromise individual agents, AiTM demonstrates how an adversary can compromise entire multi-agent systems by only manipulating the messages passing between agents. To enable the attack under the challenges of limited control and role-restricted communication format, we develop an LLM-powered adversarial agent with a reflection mechanism that generates contextually-aware malicious instructions. Our comprehensive evaluation across various frameworks, communication structures, and real-world applications demonstrates that LLM-MAS is vulnerable to communication-based attacks, highlighting the need for robust security measures in multi-agent systems.'}

Research Trends

• able

• achieve

• actor-critic

• adaptation

• additionally

• agent

• agent-in-the-middle

• agreement

• annotation

• area

• attack

• attribution

• automated

• benchmark

• better

- challenged
- communication
- communication-based
- competitive
- contextually-aware
- cooperative
- coordination
- dataset
- design
- difficulty
- failure
- failure-responsible
- failures-provide
- fine-grained
- inter-agent
- inter-annotator
- labor-intensive
- llm-as-a-judge
- llm-ma
- llm-powered
- mast
- mast-data
- message-based
- model-based
- multi-agent
- non-stationarity
- policy
- problem-solving
- q-learning
- real-world
- reasoning
- reinforcement
- role-restricted
- steps
- systematic
- systems-identifying

Research Gaps

• These results highlight the task's complexity and the need for further research in this area.

Future Scope

• These results highlight the task's complexity and the need for further research in this area.

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

• {'title': 'Consensus and Cooperation in Networked Multi-Agent Systems', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Not Available', 'methodology': ['Not Identified'], 'keywords': [], 'year': 2007, 'citation_count': 10390, 'quality_score': 50, 'quality_classification': 'Good'}

• {'title': 'Why Do Multi-Agent LLM Systems Fail?', 'research_area': 'Retrieval-Augmented Generation', 'research_problem': 'This gap highlights a critical need for a principled understanding of why MAS fail.', 'methodology': ['Framework', 'Experimental Evaluation', 'Dataset', 'Large Language Model', 'Retrieval-Augmented Generation'], 'keywords': ['multi-agent', 'mast-data', 'dataset', 'failure', 'agent', 'benchmark', 'inter-agent', 'inter-annotator', 'llm-as-a-judge', 'mast', 'annotation', 'systematic', 'agreement', 'better', 'design'], 'year': 2025, 'citation_count': 535, 'quality_score': 100, 'quality_classification': 'Excellent'}

• {'title': 'Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive.', 'methodology': ['Dataset', 'Large Language Model'], 'keywords': ['multi-agent', 'agent', 'failure', 'dataset', 'reasoning', 'attribution', 'failure-responsible', 'failures-provide', 'fine-grained', 'labor-intensive', 'systems-identifying', 'achieve', 'area', 'automated', 'steps'], 'year': 2025, 'citation_count': 141, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments', 'research_area': 'Reinforcement Learning', 'research_problem': 'We begin by analyzing the difficulty of traditional algorithms in the multi-agent case: Q-learning is challenged by an inherent non-stationarity of the environment, while policy gradient suffers from a variance that increases as the number of agents grows.', 'methodology': ['Algorithm', 'Reinforcement Learning'], 'keywords': ['multi-agent', 'agent', 'reinforcement', 'policy', 'actor-critic', 'non-stationarity', 'q-learning', 'coordination', 'able', 'adaptation', 'additionally', 'challenged', 'competitive', 'cooperative', 'difficulty'], 'year': 2017, 'citation_count': 6284, 'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'Red-Teaming LLM Multi-Agent Systems via Communication Attacks', 'research_area': 'Multi-Agent Systems', 'research_problem': 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications.', 'methodology': ['Framework', 'Experimental Evaluation', 'Large Language Model'], 'keywords': ['multi-agent', 'agent', 'communication', 'llm-ma', 'agent-in-the-middle', 'communication-based', 'contextually-aware', 'inter-agent', 'llm-powered', 'message-based', 'model-based', 'problem-solving', 'real-world', 'role-restricted', 'attack'], 'year': 2025, 'citation_count': 127, 'quality_score': 90, 'quality_classification': 'Excellent'}

Highest Cited Paper

title: Consensus and Cooperation in Networked Multi-Agent Systems

authors: ['R. Olfati-Saber', 'J. Fax', 'R. Murray']

year: 2007

citation_count: 10390

summary: Abstract not available for analysis.

research_problem: Not Available

methodology: ['Not Identified']

key_contributions: []

future_work: []

keywords: []

research_area: Multi-Agent Systems

paper_score: 50

paper_quality: Good

Latest Paper

title: Why Do Multi-Agent LLM Systems Fail?

authors: ['M. Cemri', 'Melissa Z. Pan', 'Shuyi Yang', 'Lakshya A. Agrawal', 'Bhavya Chopra', 'Rishabh Tiwari', 'Kurt Keutzer', 'Aditya G. Parameswaran', 'Dan Klein', 'K. Ramchandran', 'Matei A. Zaharia', 'Joseph E. Gonzalez', 'Ion Stoica']

year: 2025

citation_count: 535

summary: Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal. This gap highlights a critical need for a principled understanding of why MAS fail. Addressing this question requires systematic identification and analysis of failure patterns. We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks. MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems. To enable systematic classification of failures for MAST-Data, we build the first Multi-Agent System Failure Taxonomy (MAST). We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement (kappa = 0.88). This process identifies 14 unique modes, clustered into 3 categories: (i) system design issues, (ii) inter-agent misalignment, and (iii) task verification. To enable scalable annotation, we develop an LLM-as-a-Judge pipeline with high agreement with human annotations. We leverage MAST and MAST-Data to analyze failure patterns across models (GPT4, Claude 3, Qwen2.5, CodeLlama) and tasks (coding, math, general agent), demonstrating improvement headrooms from better MAS design. Our analysis provides insights revealing that identified failures require more sophisticated solutions, highlighting a clear

research_problem: This gap highlights a critical need for a principled understanding of why MAS fail.

methodology: ['Framework', 'Experimental Evaluation', 'Dataset', 'Large Language Model', 'Retrieval-Augmented Generation']

key_contributions: ['We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks.', 'We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement (kappa = 0.88).', 'To enable scalable annotation, we develop an LLM-as-a-Judge pipeline with high agreement with human annotations.']

future_work: []

keywords: ['multi-agent', 'mast-data', 'dataset', 'failure', 'agent', 'benchmark', 'inter-agent', 'inter-annotator', 'llm-as-a-judge', 'mast', 'annotation', 'systematic', 'agreement', 'better', 'design']

research_area: Retrieval-Augmented Generation

paper_score: 100

paper_quality: Excellent

Common Methodologies

• Algorithm

• Dataset

• Experimental Evaluation

• Framework

• Large Language Model

• Not Identified

• Reinforcement Learning

• Retrieval-Augmented Generation

Research Areas

• Multi-Agent Systems

• Reinforcement Learning

• Retrieval-Augmented Generation

Common Keywords

• able

• achieve

• actor-critic

• adaptation

• additionally

• agent

• agent-in-the-middle

• agreement

• annotation

• area

• attack

• attribution

• automated

• benchmark

• better

• challenged

- communication
- communication-based
- competitive
- contextually-aware
- cooperative
- coordination
- dataset
- design
- difficulty
- failure
- failure-responsible
- failures-provide
- fine-grained
- inter-agent
- inter-annotator
- labor-intensive
- llm-as-a-judge
- llm-ma
- llm-powered
- mast
- mast-data
- message-based
- model-based
- multi-agent
- non-stationarity
- policy
- problem-solving
- q-learning
- real-world
- reasoning
- reinforcement
- role-restricted
- steps
- systematic
- systems-identifying

Methodology Frequency

Large Language Model: 3

Framework: 2

Experimental Evaluation: 2

Dataset: 2

Not Identified: 1

Retrieval-Augmented Generation: 1

Algorithm: 1

Reinforcement Learning: 1

Most Common Methodology: Large Language Model

Quality Distribution

Good: 1

Excellent: 3

Very Good: 1

Publication Years

2007: 1

2017: 1

2025: 3

Citation Statistics

highest: 10390

lowest: 127

average: 3495.4

total: 17477

Comparison Summary

total_papers: 5

most_common_methodology: Large Language Model

research_areas: ['Multi-Agent Systems', 'Reinforcement Learning', 'Retrieval-Augmented Generation']

quality_distribution: {'Good': 1, 'Excellent': 3, 'Very Good': 1}

citation_statistics: {'highest': 10390, 'lowest': 127, 'average': 3495.4, 'total': 17477}

highest_cited_title: Consensus and Cooperation in Networked Multi-Agent Systems

latest_paper_title: Why Do Multi-Agent LLM Systems Fail?

Research Gap Analysis

Total Papers: 5

Research Areas

- Multi-Agent Systems
- Reinforcement Learning
- Retrieval-Augmented Generation

Common Keywords

- agent
- multi-agent
- dataset
- failure
- inter-agent
- able
- achieve
- actor-critic
- adaptation
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- agent-in-the-middle
- agreement
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• role-restricted
• steps
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• systems-identifying

Future Work

• These results highlight the task's complexity and the need for further research in this area.

Research Area Distribution

Multi-Agent Systems: 3

Retrieval-Augmented Generation: 1

Reinforcement Learning: 1

Keyword Frequency

agent: 4

multi-agent: 4

dataset: 2

failure: 2

inter-agent: 2

able: 1

achieve: 1

actor-critic: 1

adaptation: 1

additionally: 1

agent-in-the-middle: 1

agreement: 1

annotation: 1

area: 1

attack: 1

attribution: 1

automated: 1

benchmark: 1

better: 1

challenged: 1

communication: 1

communication-based: 1

competitive: 1

contextually-aware: 1

cooperative: 1

coordination: 1

design: 1

difficulty: 1

failure-responsible: 1

failures-provide: 1

fine-grained: 1

inter-annotator: 1

labor-intensive: 1

llm-as-a-judge: 1

llm-ma: 1

llm-powered: 1

mast: 1

mast-data: 1

message-based: 1

model-based: 1

non-stationarity: 1

policy: 1

problem-solving: 1

q-learning: 1

real-world: 1

reasoning: 1

reinforcement: 1

role-restricted: 1

steps: 1

systematic: 1

systems-identifying: 1

Research Trends

- agent
- multi-agent
- dataset
- failure
- inter-agent
- able
- achieve
- actor-critic
- adaptation
- additionally
- agent-in-the-middle
- agreement
- annotation
- area
- attack

Gap Categories

datasets: []

models: ['Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive.', 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications.']

evaluation: []

applications: []

coordination: ['Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive.', 'We begin by analyzing the difficulty of traditional algorithms in the multi-agent case: Q-learning is challenged by an inherent non-stationarity of the environment, while policy gradient suffers from a variance that increases as the number of agents grows.', 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications.']

security: []

scalability: []

explainability: []

general: ["These results highlight the task's complexity and the need for further research in this area.", 'Not Available', 'This gap highlights a critical need for a principled understanding of why MAS fail.']

Research Gaps

• Current models and architectures still require improvements in reliability, adaptability, and generalization.

• Reliable coordination and communication between multiple agents remains an open research challenge.

• This gap highlights a critical need for a principled understanding of why MAS fail.

• Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive.

• These results highlight the task's complexity and the need for further research in this area.

• We begin by analyzing the difficulty of traditional algorithms in the multi-agent case: Q-learning is challenged by an inherent non-stationarity of the environment, while policy gradient suffers from a variance that increases as the number of agents grows.

• Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications.

Emerging Topics

• agent

• multi-agent

• dataset

• failure

• inter-agent

Recommendations

• Explore more reliable autonomous and multi-agent systems.

Summary

dominant_area: Multi-Agent Systems

top_trend: agent

future_work_items: 1

research_gap_count: 7

Experiment Plan

Research Areas

• Multi-Agent Systems

• Reinforcement Learning

• Retrieval-Augmented Generation

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

• Accuracy
• Precision
• Recall
• F1-Score
• ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

• Train-Test Split
• K-Fold Cross Validation
• Hyperparameter Tuning
• Statistical Significance Testing

Experimental Workflow

• Collect Dataset
• Preprocess Data
• Train Baseline Models
• Train Proposed Model
• Evaluate Performance
• Compare Results
• Analyze Errors
• Draw Conclusions

Report Summary

Dominant Research Area: Multi-Agent Systems

Top Research Trend: agent

Highest Cited Paper: Consensus and Cooperation in Networked Multi-Agent Systems

Latest Paper: Why Do Multi-Agent LLM Systems Fail?

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.